ES 404: Networks and Complex Systems

Assignment 3 — Spreading over a Campus Network

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1. Network Setup

1.1. Campus Zones and Population

The campus is partitioned into nine zones, each represented as a node in a weighted contact graph. Table 1 summarises zone meanings and resident populations.

Table 1: Campus zones and populations.

Node	Zone	Population N_i
H1	Hostel cluster 1	320
H2	Hostel cluster 2	280
H3	Hostel cluster 3	260
D	Dining hall / mess	450
L	Library / common study	180
A	Academic core	520
Lab	Labs / workshops	240
S	Sports / student activity	160
G	Main gate / bus stop	120
	Total	**2530**

1.2. Contact Network

Fourteen undirected weighted edges encode pairwise contact intensity between zones. Edge weight is a dimensionless relative contact rate; a weight of 1.0 (D – A) reflects the highest-frequency inter-zone interaction on campus. Table 2 lists all edges; Figure 1 visualises the graph with node size proportional to population and edge width proportional to weight.

Table 2: Campus contact edges and weights.

From	To	Weight
H1	D	0.8
H2	D	0.9
H3	D	0.7
H1	A	0.4
H2	A	0.5
H3	A	0.3
D	A	1.0
D	L	0.5
D	S	0.4
A	L	0.7
A	Lab	0.8
A	G	0.4
Lab	G	0.3
L	G	0.2

1.3. Structural Analysis and Outbreak Prediction

(b) The graph structure predicts the outbreak's shape before any simulation begins.

The **academic core (A)** is the hub. With seven edges and a summed contact weight of 4.1, it

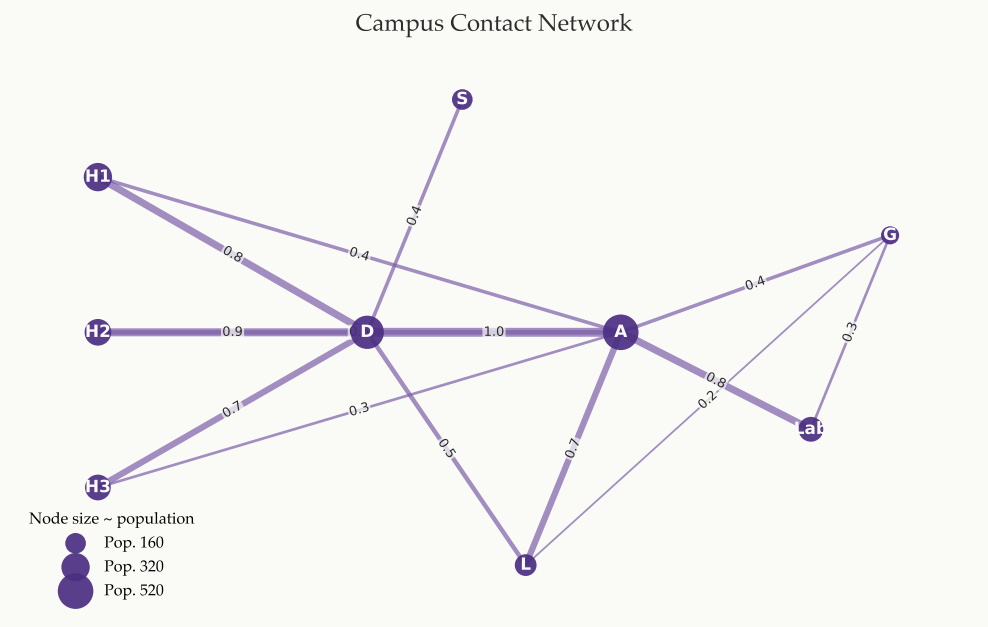


Figure 1: Campus contact network. Node size scales with zone population; edge width scales with contact weight. The academic core (A) has the highest weighted degree ($\sum w = 4.1$), while Sports (S) is a structural leaf with a single link.

connects hostels, dining hall, library, labs, and gate simultaneously. The D–A edge (weight 1.0) is the strongest on campus, making this corridor the epidemic's likely engine. Any seed reaching A will radiate outward in all directions at once.

The **library (L)** is the bridge. It is the only path from the hostel-dining cluster to the gate node (via A–L–G). Remove L and G detaches epidemiologically; keep it open and it becomes a conduit for late-arriving infection.

The **sports area (S)** is the peripheral node. Its single edge to D (weight 0.4) is a structural dead-end: infection enters but cannot propagate. S will be infected last and least.

Prediction: a seed in H2 will traverse the H2–D and H2–A edges and ignite a campus-wide wave within five to seven days. A seed in A will do the same, faster, because it skips that relay entirely. G and S will lag the rest of campus in both cases, insulated by weak and sparse connections.

2. Model and Simulation

2.1. Zone-Level SIR Model

Each zone i evolves under a modified SIR model where the force of infection $\lambda_i(t)$ combines local within-zone prevalence and a network-mediated import term:

$$\lambda_i(t) \;=\; \beta\left[\rho\,\frac{I_i}{N_i} \;+\; (1-\rho)\sum_j \widehat{W}_{ij}\,\frac{I_j}{N_j}\right] \tag{1}$$

$$\begin{aligned}\frac{dS_i}{dt} &= -\lambda_i\, S_i,\\ \frac{dI_i}{dt} &= \lambda_i\, S_i - \mu\, I_i,\\ \frac{dR_i}{dt} &= \mu\, I_i.\end{aligned}$$

Here $\widehat{W}_{ij}$ is the *row-normalised* adjacency matrix: each row sums to unity so the network term is a proper weighted average of neighbouring prevalences. Population in each zone is conserved: $N_i = S_i + I_i + R_i = \text{const}$.

Parameters: $\beta = 0.34\,\text{day}^{-1}$, $\mu = 0.20\,\text{day}^{-1}$ (infectious period ≈ 5 days), $\rho = 0.55$ (55% local, 45% network-driven transmission), horizon $= 30$ days.

The system of 27 coupled ODEs is integrated with `scipy.integrate.solve_ivp` using `RK45`, `rtol=1e-8`, `atol=1e-10`, and dense output. The solver conserves total population to machine precision (verified: $\sum S_i + I_i + R_i = 2530$ at all times).

2.2. Seed Scenarios

(c) Two initial conditions are considered, with $R_i(0) = 0$ and $S_i(0) = N_i - I_i(0)$ for all zones:

Scenario A $I_{H2}(0) = 5$, all other zones zero. The outbreak begins in Hostel 2, a well-connected node with strong links to both D (0.9) and A (0.5).

Scenario B $I_A(0) = 5$, all other zones zero. The outbreak begins in the academic core, the network's highest-degree hub.

3. Simulation Results

3.1. Total Campus Infection Dynamics

(d) Figure 2 shows total infected population across campus over the 30-day horizon for both seed scenarios. In both cases, the epidemic is still in its growth phase at day 30 — consistent with an $R_0 = \beta/\mu = 1.70$ outbreak seeded in only $5/2530 \approx 0.2\%$ of the population, which needs time to build momentum. By day 30, Scenario A accumulates **159** concurrent infections (446 ever infected), while Scenario B reaches **204** (609 ever infected), confirming that hub seeding produces a materially faster and larger outbreak.

3.2. Per-Zone Infection Curves

(e) Figure 3 disaggregates the outbreak into four key zones: H2 (the Scenario A seed), A (the hub / Scenario B seed), D (dining hall, the strongest single edge), and G (the peripheral gate node).

In Scenario A the infection emerges from H2 and flows first to D (strong link 0.9) and then outward through A to the rest of campus. The academic core and dining hall quickly overtake the

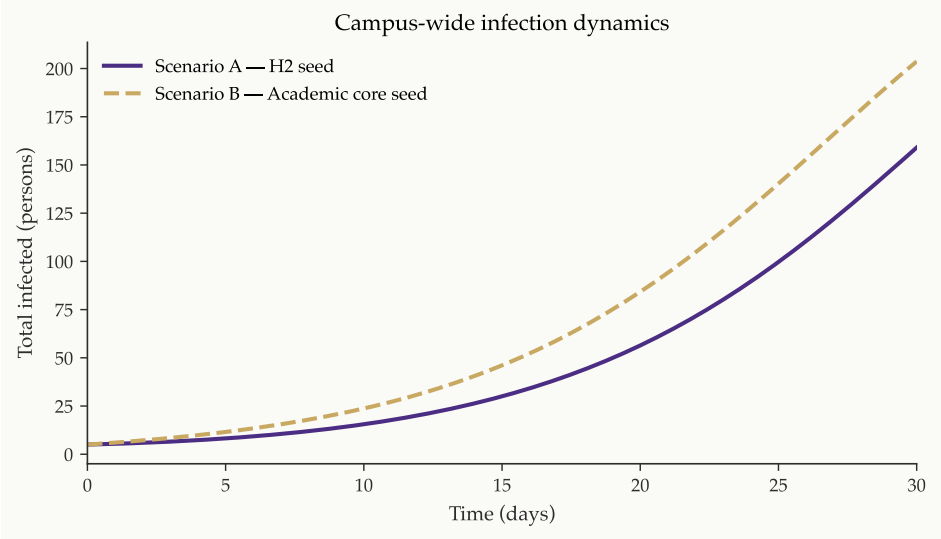


Figure 2: Total infected population versus time for both seed scenarios. Scenario B (academic core seed, dashed gold) grows faster and higher than Scenario A (H2 hostel seed, solid purple) throughout the 30-day window.

hostel seed in absolute infected count because their larger populations provide more susceptibles. G lags all others, reaching barely 7 infected by day 30 — the library-gate bottleneck slows its exposure.

In Scenario B the roles reverse: A climbs fastest from its seeded start, simultaneously lighting up D, Lab, and the three hostels. H2 is now a recipient rather than a source, yet still receives infection faster than in Scenario A because A delivers it directly (weight 0.5) rather than via the D relay.

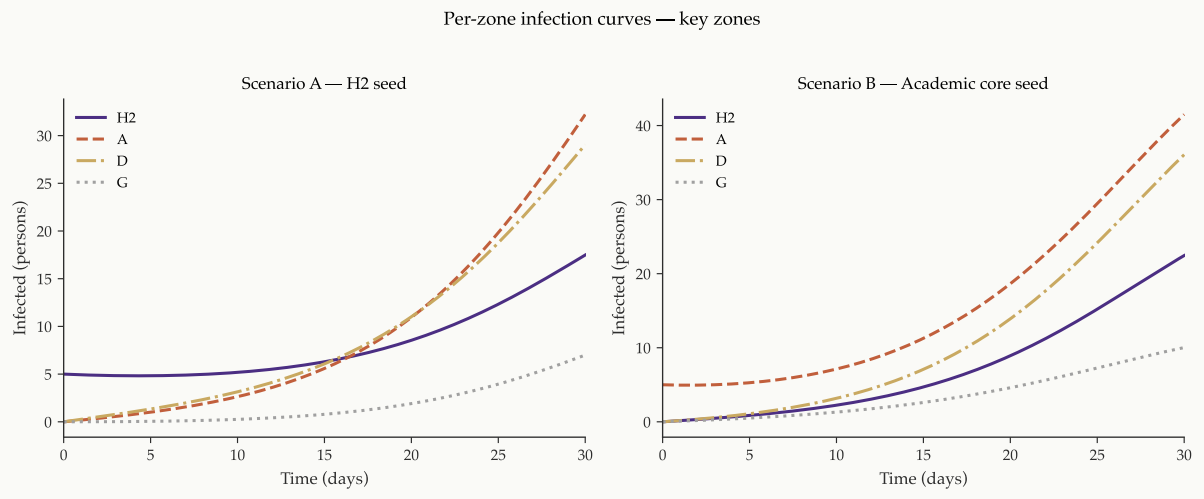


Figure 3: Infected count over time in four key zones. *Left*: Scenario A (H2 seed); *right*: Scenario B (academic core seed). Line style differentiates zones. Note how G lags all other zones in both scenarios.

3.3. Spatial Spread Snapshots

(f) Figures 4 and 5 show network snapshots at days 0, 5, 10, and 20, with node colour encoding infected fraction (white = 0, deep purple = maximum). The shared colour scale across all panels

makes temporal and cross-scenario comparison direct.

In Scenario A (Figure 4), the infection radiates outward from H2: by day 5 it has reached D and A; by day 10 the academic cluster is visible; by day 20 every node shows measurable colour, but G and S remain pale. The spread follows the network skeleton, not Euclidean distance.

In Scenario B (Figure 5), the academic core is dark from day 5 and the entire network is noticeably infected by day 10 — a full snapshot earlier than Scenario A. The hub advantage is visible at a glance.

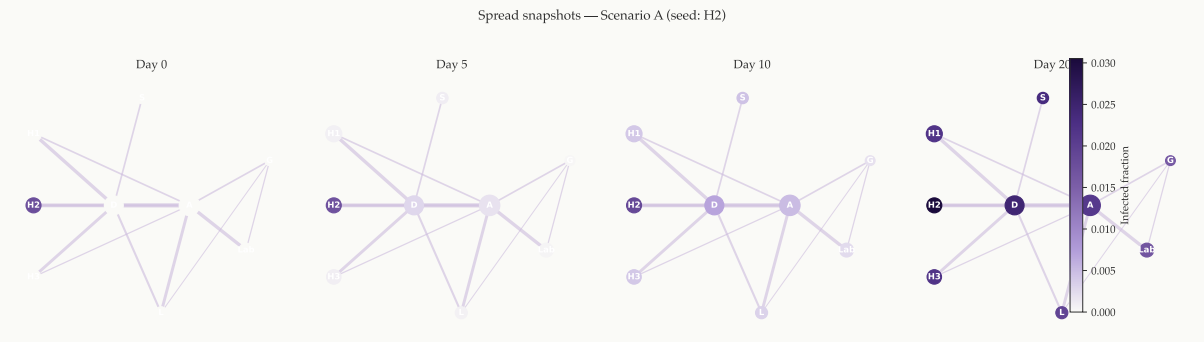


Figure 4: Snapshot of campus infection state at days 0, 5, 10, 20 — **Scenario A** (H2 seed). Node colour encodes infected fraction; shared colour scale across all panels.

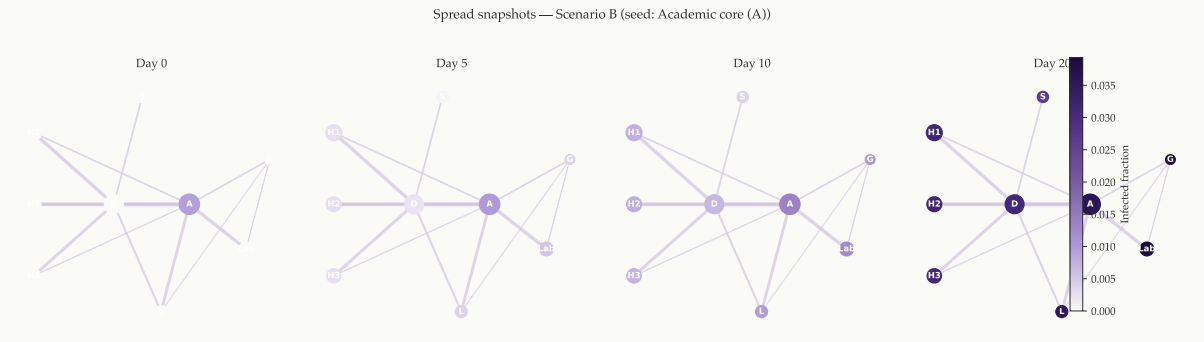


Figure 5: Same snapshot sequence for **Scenario B** (academic core seed). The campus darkens a full snapshot sooner than Scenario A, reflecting the hub's superior seeding position.

3.4. Comparison of Seed Scenarios

(g) The two scenarios differ not just in speed but in the *sequence* with which zones are recruited into the outbreak.

In Scenario A, the virus must first traverse the H2–D or H2–A edges before it can access the hub's full connectivity. This one-hop delay costs approximately 3–4 days of exponential growth. By the time A is seeded indirectly (around day 4–5), Scenario B has already radiated outward through all seven of A's neighbours. The cumulative difference is 45 additional infected people by day 30.

Zone-ordering tells the same story topologically. In Scenario A, the sequence is H2 $\rightarrow$ D $\rightarrow$ A $\rightarrow$ {H1, H3, L, Lab} $\rightarrow$ {S, G}. In Scenario B it collapses to A $\rightarrow$ {D, H1, H2, H3, L, Lab, G} in near-parallel, because A is directly linked to all of them. The peripheral nodes S and G are last in both scenarios for the same structural reason: S has degree one (only D), and G is only

reachable via the weak L–G (0.2) or Lab–G (0.3) edges.

The broader lesson is that *proximity to the hub matters more than proximity to the seed*. A node two hops from A but strongly connected to D (e.g. H1) becomes infected faster than a node one hop from H2 but weakly connected overall (e.g. H3 via D).

4. Well-Mixed SIR Baseline Comparison

(h) Figure 6 compares the total network-infected curve for Scenario B against a well-mixed SIR model with identical parameters: $\beta = 0.34$, $\mu = 0.20$, total population $N = 2530$, $I(0) = 5$.

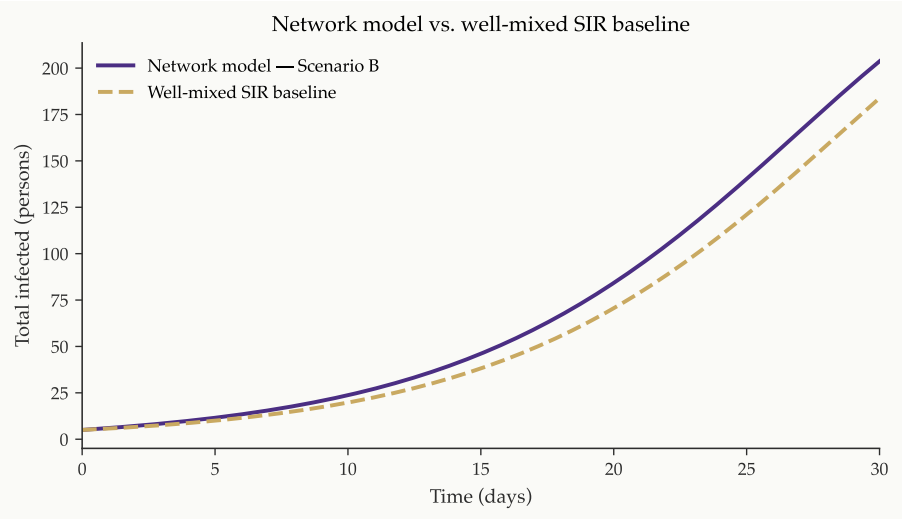


Figure 6: Network model (Scenario B, solid purple) versus the well-mixed SIR baseline (dashed gold). Both share identical β, μ, and $I(0) = 5$. The network model accumulates infections faster when the hub is seeded.

What does the well-mixed model miss? By day 30 the network model produces 204 concurrent infections against the well-mixed baseline's 183 — an 11% gap that widens with time.

The well-mixed model assumes that every infected person contacts every susceptible at the same rate. In reality, a virus seeded in the academic core immediately benefits from A's seven high-weight edges: it floods the dining hall (weight 1.0), the labs (0.8), the library (0.7), and three hostels simultaneously. The early-exponential advantage of hitting the hub is invisible to the well-mixed model, which sees only aggregate prevalence.

The well-mixed model also cannot capture the *last-mile problem*: G and S are structurally protected by their peripheral position regardless of how much infection circulates elsewhere. A well-mixed model sees a single rising pool; the network model sees pockets of protection persisting deep into the outbreak.

In short, the well-mixed baseline both *underestimates early spread* when the hub is seeded and *overestimates late-stage exposure* of structurally shielded nodes. These are not small corrections — they are qualitatively different stories about who gets infected when.

5. Intervention: Isolating the H2 Hostel Cluster

(i) Starting on day 7 — one week after the first case appears — all edges touching Hostel 2 are reduced by 80% (multiplied by a factor of 0.2), and the weight matrix is re-normalised. This mimics a quarantine of patient zero's hostel: residents still have some contact (residual 20%), but the primary H2–D (0.9) and H2–A (0.5) transmission channels are severely curtailed.

The intervention is implemented as a two-segment ODE solve: the system runs on the original weight matrix from day 0 to day 7, then restarts from that state on the modified matrix for days 7–30. This ensures exact continuity of the state vector at the changepoint.

Figure 7 shows the result.

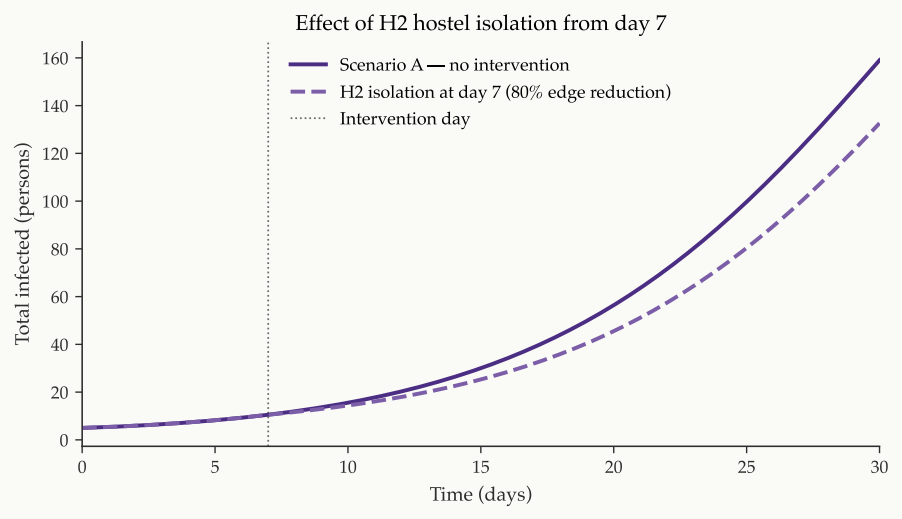


Figure 7: Scenario A baseline (solid purple) versus H2 isolation from day 7 (dashed lavender). The vertical dotted line marks the intervention. By day 30 the intervention reduces concurrent infections from 159 to 133 (16.6% reduction) and cumulative ever-infected from 446 to 371 (17% reduction).

Interpretation. The intervention acts primarily through *peak reduction* rather than meaningful peak delay: by day 7 infection has already seeded A and D via H2's strong links, so the outbreak cannot be stopped — only slowed. The curves diverge immediately after day 7, growing at visibly different rates because H2's contribution to network-mediated transmission ($\approx$ 45% network term, $\rho = 0.55$) is sharply cut.

The 16.6% reduction in concurrent infections and 17% reduction in cumulative cases after a single hostel isolation is a meaningful result: it delays the crossing of any given burden threshold by several days, buying time for clinical capacity. Had the intervention been applied at day 0 (before the virus left H2), the effect would be far larger; the lesson is that contact network structure dictates both the leverage point *and* the deadline for intervention.

6. Reflection

What did the network structure reveal that the equations alone would not have shown?

The equations describe what happens to populations; the network describes *to whom*. Without the graph, the SIR model offers a clean but fictional story: a single rising wave, a single peak, a single herd-immunity threshold. With the graph, the fiction dissolves.

The most striking thing the network revealed was *sequence*. The same pathogen, with identical β and μ, tells two completely different stories depending on where it starts. A seed in the academic core races outward in all directions at once; a seed in a hostel must find its way to the hub before it can do the same damage. The equations would produce the same $R_0 = 1.70$ either way and make no prediction about which zones fall first.

The network also revealed *structural immunity*: the sports field and main gate remain lightly infected throughout, not because they are biologically different, but because the graph gives them only one weak entry point each. No amount of staring at the differential equations would reveal this — it is a topological fact, not a dynamical one.

Finally, the intervention analysis showed that the network structure defines both the *leverage point* and the *expiry date* of any containment strategy. Isolating H2 after the virus has already jumped to A and D yields only a 17% reduction; isolating it before that jump would have been transformative. The equations give you the recovery rate; the network tells you when it is already too late.

The network structure, in short, is not a decoration on the model. It is the model.

All simulations performed in Python 3 using `numpy`*,* `scipy` *(*`solve_ivp`*, RK45, rtol* $= 10^{-8}$*),* `networkx`*, and* `matplotlib`*. Source code available in the* `src/` *directory.*